# Programming Fundamentals II: Lab 03

**Due Date: Tuesday, September 24, 2024 at Midnight**

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**Part 1. Fairy Tale Analysis (10 Marks)**

For this lab, we are going to do some basic processing on a text. Download the Lab03.java and fairytale.txt file provided. The fairytale.txt file contains the text of a German Fairy Tale. For this lab we are going to determine a few stats about this story. Using the readFile method provided, read the Fairy Tale into a String variable, then solve the following problems:

a. **(6 Marks)** Break down your story String into a 2-Dimensional array of words. In this array, the first dimension represents sentences, while the second dimension represents individual words. Use the **.split()** method of the String class to break up your text.

   a. We define a sentence here as a group of characters delimited by one or more periods, question marks, or exclamation points, which are then followed by one or more whitespace characters.
   b. We define a word here as a group of characters delimited by one or more white spaces characters or dashes.

      *In Regex, the pattern "`\\s`" matches any whitespace.*
      *Remember: Periods are special characters in Regex!*

b. **(3 Marks)** Display the following stats about the length of your story:

   a. How many characters long is the story?
   b. How many words long is the story?
   c. How many sentences long is the story?

c. **(1 Mark)** Modify the readFile() method so that it will raise an IllegalArgumentException if the fileName that is passed in is not a txt file. That is, the file name does not end in ".txt". Use the **.matches()** method of the String class. This regex pattern will return true if a file ends in ".txt":

```
.*\\.txt$
```

Add the **date and your name and student number** to the top of the java file. Please submit your.java file to Nexus before the end of the day on the due date. **Late submissions will be not be accepted.**

Once stored into our array, the story may look something like this if we were to print out each of the sentences one at a time:

```
[Es, war, einmal, ein, Fischer, und, seine, Frau,, die, wohnten,
zusammen, in, einer, kleinen, Fischerhütte,, dicht, an, der, See,, und,
der, Fischer, ging, alle, Tage, hin, und, angelte:, und, angelte, und,
angelte]
[So, saß, er, auch, einmal, mit, seiner, Angel, und, sah, immer, in,
das, klare, Wasser, hinein:, und, so, saß, er, nun, und, saß]
[Da, ging, die, Angel, auf, den, Grund,, tief, hinunter,, und, als, er,
sie, heraufhohe,, da, holte, er, einen, großen, Butt, heraus]
[Da, sagte, der, Butt, zu, ihm:, "Hör, mal,, Fischer,, ich, bitte,
dich,, laß, mich, leben,, ich, bin, kein, richtiger, Butt,, ich, bin,
ein, verwunschener, Prinz]
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